

The reasoning-effort dial stops paying long before it reaches the top.

Claude Fable 5.1 across its five effort settings, scored for coding

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Reasoning effort is not a quality dial; it is a budget for deliberation over whatever is already in the context window. That distinction decides the whole analysis. Raising effort lets the model spend more tokens reasoning before it commits to an answer, which helps when the answer is genuinely hard to derive from material the model can already see. It does nothing at all when the material in front of the model is the wrong material. Fable 5.1’s five settings are therefore best read as five points on a deliberation-budget curve, and the interesting question is where that curve stops repaying its cost.

1 What the dial actually controls

Fable 5.1 ships with five effort settings: low, medium, high, xhigh and max. Anthropic sets high as the default in Claude Code and medium in the consumer surfaces, and states that low and medium effort on 5.1 already match full-effort Fable 5. Artificial Analysis evaluated all five before release, running each through the same Intelligence Index v4.2 suite, which is the only public dataset that holds the workload constant while varying only the effort parameter.

Holding the workload constant matters more than it appears. Every published comparison of effort settings measures how much deliberation the model spends on a fixed set of tasks, not how well it performs on tasks of different shapes. The curve below therefore measures one thing precisely: the return on additional deliberation, given a fixed and correctly specified problem. It is silent on the failure mode that dominates large real codebases, which is the problem being specified from the wrong files.

2 Intelligence bought against tokens spent

The horizontal axis below is logarithmic. That choice is not cosmetic: on a log axis a constant return per doubling of spend appears as a straight line, so any downward bend is a real diminishing return rather than an artefact of the scale. Fable 5.1 bends almost immediately.

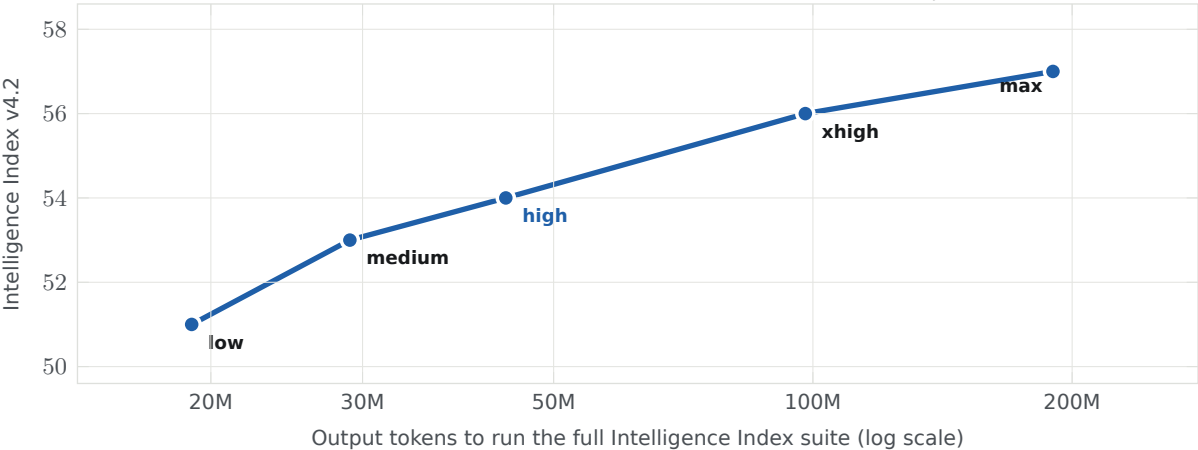


Figure 1. The effort frontier. Each point is one effort setting running the identical Intelligence Index v4.2 suite. Six index points separate the cheapest setting from the most expensive; ten times the output tokens separate them as well.

The shape is the familiar one for any search process with a finite answer space. Early deliberation eliminates the cheap errors, which are numerous; later deliberation works on the residue,

which is both smaller and harder. What varies between models is where the elbow sits. On Fable 5.1 it sits at high, and everything above high is buying the residue.

3 Coding is where the curve dies earliest

Terminal-Bench v2.1 is the agentic-coding component of the index: long-horizon terminal work, and the closest public proxy for the sort of task a coding agent is actually given. Artificial Analysis publishes it for the top three effort settings only.

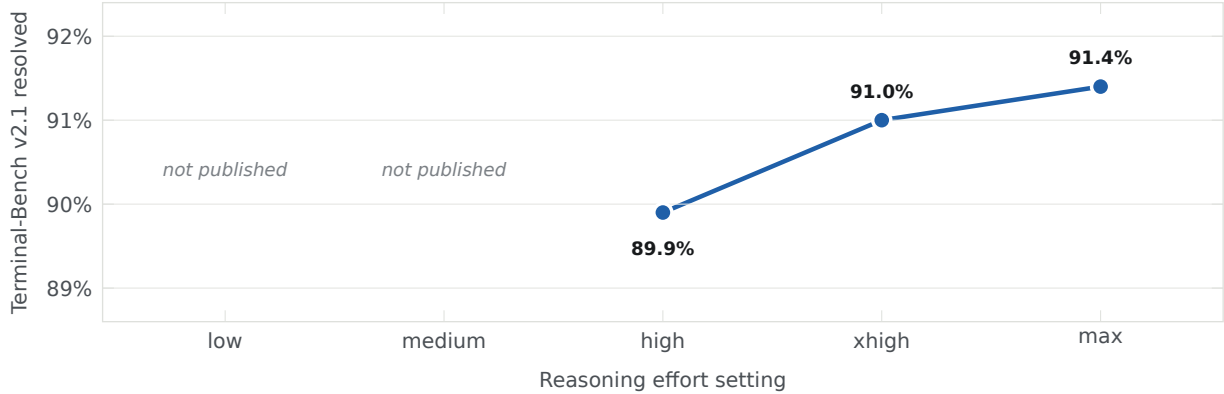


Figure 2. Agentic coding across the top three settings. The entire published spread is 3.8 percentage points wide and the three points occupy 1.5 of them. Position rather than length is used, because bars on this range would exaggerate a small difference.

Read as scores, 89.9, 91.0 and 91.4 look like a modest but real progression. Read as an error budget the picture is blunter. At high the model fails 10.1 tasks in every hundred; at max it fails 8.6. Spending 4.3 times the tokens removes 1.5 failures per hundred and leaves the other 8.6 exactly where they were. Those remaining failures are not failures of insufficient deliberation, or more deliberation would have cleared them.

This is also the regime where benchmark saturation starts to matter. A 1.5-point spread at the top of a benchmark is close to the resolution of the benchmark itself. The published numbers can tell us that max is not much better than high; they are too coarse to tell us confidently that it is better at all.

4 Marginal return collapses eighteenfold

Differencing the frontier gives the quantity that should drive the decision: index points gained per additional million output tokens spent on each step up.

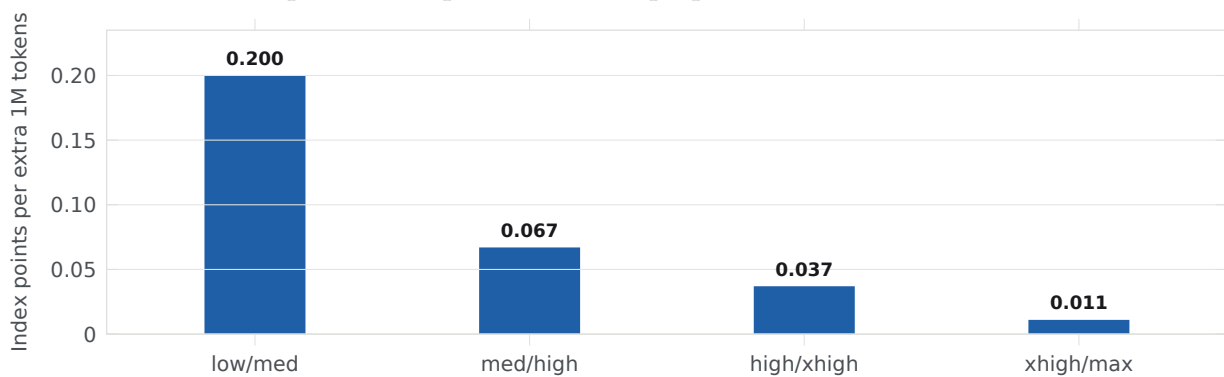


Figure 3. The value of the marginal token, by step. The last rung of the ladder is roughly eighteen times worse value per token than the first. Nothing about the model degrades; the evaluation runs out of headroom.

High is the last setting before this figure falls below 0.04. In absolute terms the same collapse is visible as efficiency: 2.68 index points per million tokens at low, 1.23 at high, 0.30 at max. High

delivers 98 per cent of max-effort Terminal-Bench performance for 23 per cent of the output tokens, and that single ratio carries most of the practical argument.

5 Latency is the cost that scales worst

Tokens are the cost that appears on an invoice. Time to first token is the cost that is felt, and it scales far worse than price does. Output speed actually rises with effort, from 52.8 to 69.5 tokens per second, which is why cost per task grows by only 3.6 times across the range while the wait before anything appears grows by 47 times.

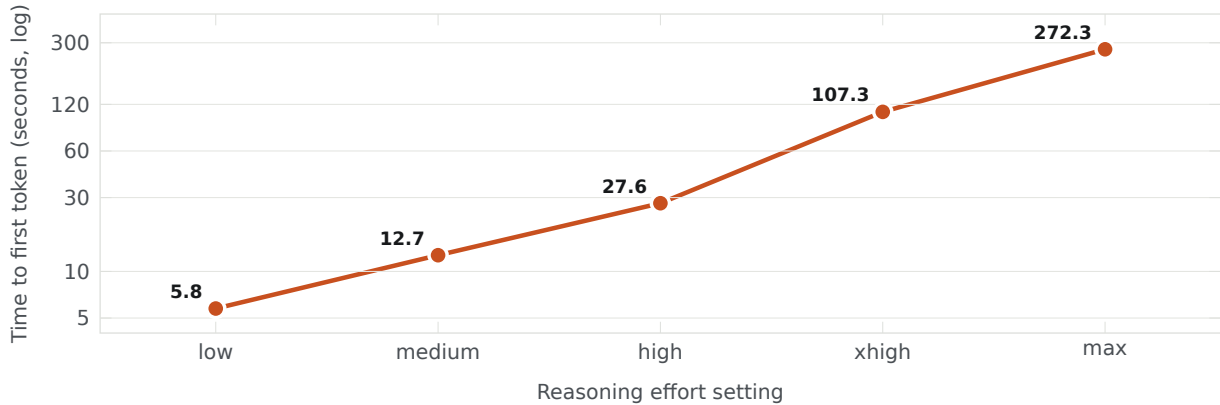


Figure 4. The latency tax. At max effort the median first token arrives after four and a half minutes. For an interactive editing loop this is usually the binding constraint well before spend is.

The mechanism is straightforward: effort budgets are spent before the first output token is emitted, so the entire deliberation cost is front-loaded into the wait. A four-minute pause does not merely cost four minutes. It breaks the loop in which a developer holds a hypothesis in working memory while the agent tests it, and that loop is most of the value of an interactive coding agent.

6 The full table

Effort	Index	T-Bench	Tokens	\$/task	TTFT	pts/M
low	51	–	19M	\$1.70	5.8s	2.68
medium	53	–	29M	\$2.14	12.7s	1.83
high	54	89.9%	44M	\$2.86	27.6s	1.23
xhigh	56	91.0%	98M	\$4.64	107.3s	0.57
max	57	91.4%	190M	\$6.12	272.3s	0.30

Output tokens and cost are for running the full Intelligence Index suite, as measured by Artificial Analysis on the Adaptive Reasoning, Default Fallback variants. Pricing is \$10 per million input tokens and \$50 per million output tokens, with cache reads at \$0.25 per million.

7 What follows in practice

- **High is the defensible default**, and is already the default in Claude Code. It sits at the elbow of the curve, and the 98-per-cent-for-23-per-cent trade is the whole argument for it.
- **Medium deserves more consideration than it gets.** If the parity claim holds, medium on 5.1 is full-effort Fable 5 at 12.7 seconds to first token rather than 27.6. For scaffolding, test writing and mechanical refactors, the halved latency is worth more than two index points.
- **Xhigh is a per-task escalation, not a session default.** 1.1 percentage points of Terminal-Bench for 2.2 times the tokens is a poor standing arrangement and a reasonable one-off.
- **Max is hard to justify for coding.** 0.4 points over xhigh for 1.9 times the tokens and 2.5 times

the wait. Its case is single-shot research-grade reasoning, not an editing loop.

8 What this evidence cannot settle

- **Two sets of numbers from the same source disagree.** The Artificial Analysis launch article quotes an index range of 58 to 66 across 13.1M to 143.7M output tokens; the per-model pages and release comparison table give 51 to 57 across 19M to 190M. This analysis uses the model pages throughout, because they are internally consistent across all five settings and are the only ones carrying per-setting cost and latency. The shape of the curve is the same on either set; the absolute levels are not settled.
- **Low and medium have no published Terminal-Bench score,** so the coding curve rests on three points rather than five, and the two cheapest settings are inferred from the composite index alone.
- **The larger limitation is one of construct.** These are eval-suite aggregates over tasks whose context is supplied correctly by the harness. On a codebase of any real size the dominant failure mode is not shallow reasoning over the right files; it is confident reasoning over the wrong ones. Raising effort makes that failure worse rather than better, because it produces more elaborate and more plausible inference from the same faulty premises.

Sources. Artificial Analysis: Claude Fable 5.1 model pages for low, medium, high, xhigh and max effort; the release comparison table; the Terminal-Bench v2.1 leaderboard; and the article ``Claude Fable 5.1 tops the Artificial Analysis Intelligence Index''. Anthropic: ``Introducing Claude Fable 5.1 and Claude Mythos 5.1'', September 2026. All retrieved 7 September 2026. Every derived quantity here is computed from the table in section six rather than taken from a source. AI assistance was used for data retrieval, computation, figure generation, drafting and translation; no benchmark was run independently.